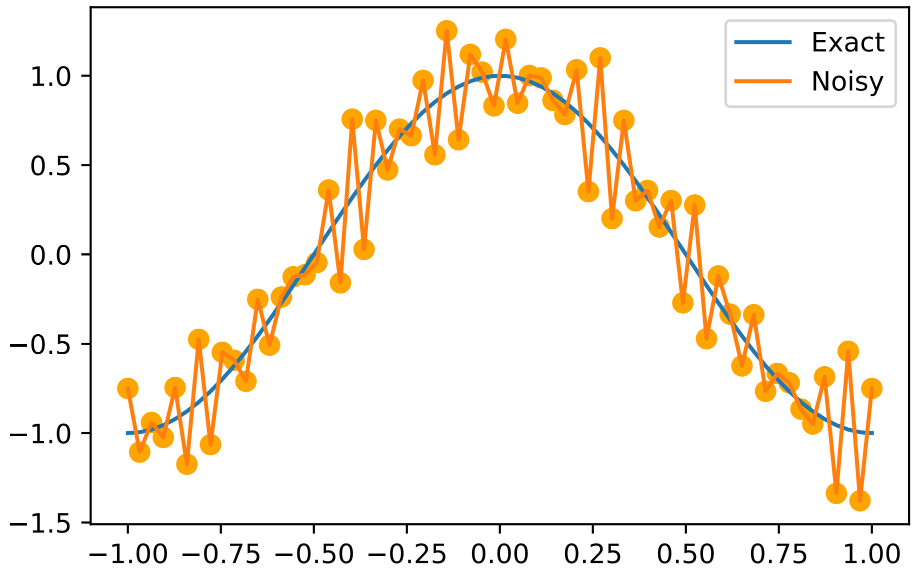
Plots for Lecture 3

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(start=-1,stop=1,num=64)
def f(x):
    return np.cos(np.pi*x)
def g(x):
    return np.cos(np.pi*x) + 1/5*np.sin(24*np.pi*x) + 1/4*np.cos(30*np.pi*x)

plt.plot(x, f(x), label='Exact')
plt.plot(x, g(x), label='Noisy')
plt.scatter(x, g(x), color='orange', s=50)  # Add dots for 'Noisy'
plt.legend()
```



```
y = np.zeros_like(x)
for k in range(1, 63):
    y[k] = 1/4*g(x[k+1]) + 1/2*g(x[k]) + 1/4*g(x[k-1])
plt.plot(x, f(x), label='Exact')
plt.plot(x, g(x), label='Noisy', alpha=0.5)
plt.scatter(x, g(x), color='orange', s=50, alpha=0.5)  # Add dots for 'Noisy'
plt.plot(x[1:-1], y[1:-1], label='Filtered')
plt.scatter(x[1:-1], y[1:-1], color='green', s=50)  # Add dots for 'Filtered'
plt.legend()
```